

HALF ADDER, FULL ADDER, HALF SUBTRACTOR, FULL SUBTRACTOR

1. What is a half adder?
 2. Why is it called "half" adder?
 3. What are its inputs and outputs?
 4. Write the truth table of a half adder.
 5. Derive the Boolean expressions for Sum and Carry.
 6. Why is XOR used for Sum?
 7. Why is AND used for Carry?
 8. Can we build a half adder using only NAND gates? How?
 9. Can half adder add 3 bits? Why not?
 10. Why is half adder not sufficient for multi-bit addition?
 11. Where is half adder used inside a full adder?
 12. Gate count comparison: XOR+AND vs only NAND implementation?
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◆ FULL ADDER – Interview Questions

◆ Basic Level

1. What is a full adder?
2. Difference between half adder and full adder?
3. Write the truth table of a full adder.
4. Derive Sum and Carry using K-map.
5. Write canonical SOP form.

◆ Intermediate (Understanding Depth)

6. Why do we need Carry-in?
7. How can you implement full adder using two half adders?
8. Derive Carry expression:

$$C_{out} = AB + BC_{in} + AC_{in}$$

9. Why is Carry called majority function?

◆ Advanced (Interviewer Favorite)

10. How many gates are required for full adder?
11. Implement full adder using only NAND gates.
12. How does propagation delay affect ripple carry adder?
13. What is critical path in ripple carry adder?

14. How can you optimize carry delay?
(Hint: Carry Lookahead Adder)

◆ Design Thinking

15. Write Verilog code for full adder.
 16. What happens if Cin is stuck-at-0 fault?
 17. How would you test full adder in hardware?
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◆ HALF SUBTRACTOR – Interview Questions

◆ Basic

1. What is a half subtractor?
2. Write truth table.
3. Derive Difference and Borrow equations.
4. Why is XOR used for Difference?
5. Why is Borrow = A'B ?

◆ Conceptual

6. What is the difference between Carry and Borrow?
 7. Can half subtractor subtract 3 bits?
 8. Why is it called "half"?
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◆ FULL SUBTRACTOR – Interview Questions

◆ Basic

1. What is a full subtractor?
2. Difference between half and full subtractor?
3. Write truth table.
4. Derive Borrow equation using K-map.

◆ Intermediate

5. Implement full subtractor using two half subtractors.
6. What is Borrow-in?
7. Write the Boolean expression for Borrow-out:

$$Bout = A'B + BBin + A'Bin$$

◆ Advanced / VLSI Level

8. Implement full subtractor using only NAND gates.
9. Compare full adder and full subtractor structurally.

10. How can you design a single circuit that performs both addition and subtraction?
(Hint: Adder-Subtractor using XOR control)
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 High-Level Conceptual Questions (To Check Real Understanding)

1. Is subtraction possible using only adders?
 2. How does 2's complement help in subtraction?
 3. Why are adders more commonly implemented than subtractors in processors?
 4. What arithmetic unit uses full adders internally?
(Hint: Arithmetic Logic Unit)
 5. Why ripple carry adder is slow?
 6. Compare ripple carry vs carry lookahead adder.
 7. What happens to delay as bit-width increases?
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 If Interviewer Wants to Test Deep Knowledge

They may ask:

- Draw transistor-level full adder.
 - CMOS implementation of XOR.
 - Power consumption difference between XOR-heavy vs NAND-only design.
 - How to reduce switching power?
 - How does fan-out affect delay?
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1. Why can't we use an OR gate instead of an XOR gate for the Sum output in a half adder?
 2. What is the main limitation of a half adder?
 3. If we connect two half adders directly, will we get a full adder? Why or why not?
 4. What happens if both inputs of a half adder toggle simultaneously?
 5. Is a half adder a combinational circuit or a sequential circuit? Why?
 6. Can we implement a half adder using only NOR gates?
 7. What determines the propagation delay of a half adder?
 8. Why is the XOR gate considered costly in VLSI implementation?
 9. Where is a half adder practically used in digital systems?
 10. If the carry output is always ignored, can we remove the AND gate from a half adder? What will the circuit behave as?
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